

Comparative Study of Theoretical and Market Option Prices Using the Black–Scholes Model

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1 Introduction

1.1 Title:

Fitting the Black-Scholes Option Pricing Model Using Historical Option Data: A Study from May to July.

1.2 Objective:

To apply and evaluate the Black-Scholes option pricing model using historical data of a stock and its associated call option contract expiring on 31st July. The project focuses on estimating key parameters—especially market volatility—and comparing the Black-Scholes theoretical price with the market-settled price to assess pricing accuracy.

1.3 Data Overview

For this analysis, I have obtained call and put option data corresponding to two different strike prices (which are closest to the spot price of the underlying around 2nd May) for each of the following four companies: INFY, HDFC, HINDALCO, and VOLTAS. In the report, I present the implementation code specifically for the INFY Call option with a strike price of 1520. The results for all other companies and strike prices are also included for completeness and comparison.

INFY CE (1520): This dataset contains historical data of option pricing which will expire on 31st July. The dataset has several variables, which are as follows:

- **Symbol:** INFY (Infosys stock).
- **Time Frame:** Option data: 2nd May to 1st July and Stock data: 1st February to 2nd July (for historical volatility estimation).
- **Expiry:** 31st July (same for all options in this dataset).
- **Option Type:** Call option.
- **Strike price (k):** 1520 (closest to the spot price of the underlying around 2nd May).
- **Open Price:** Not used due to frequent missing values.
- **High / Low / Close / LTP:-** Available but not used for direct comparison.
- **Settle Price:** Used to compare with Black-Scholes model output.
- **No. of Contracts, Turnover, Premium Turnover, Open Interest, Change in OI:** Additional descriptive stats.
- **Underlying Value:-** Daily stock closing price of INFY, used as the spot price.

The link of the merged data is here: [Option pricing data merged](#)

The link of the original data :

- [Option data](#)
- [Stock data](#)

1.4 Methodology:

1.4.1 Step 1: Data Preparation

- Merge the stock data (underlying INFY prices) and option data by date.

1.4.2 Step 2: Estimation of Parameters

- **Risk-Free Rate (r):** Fixed at 6% annualized.
- **Strike Price (K):** 1520 (from option data).
- **Spot Price (S):** Daily closing price of INFY from 2nd May to 1st July.
- **Time to Maturity (T):**

$$T = \frac{\text{Months to Expiry}}{12}$$

- **Volatility (σ):** Estimated volatility using data from 1st Feb to 2nd May. Rolling window volatility can also be calculated to update σ each day.

1.4.3 Step 3: Compute Theoretical Option Prices

- Use Black-Scholes formula for a European call option:

$$C = \frac{S_0\Phi(\omega) - ke^{-rt}\Phi(\omega - \sigma\sqrt{t})}{\sigma\sqrt{t}}$$

Where

$$\omega = \frac{rt + \sigma^2 t/2 - \log(k/S_0)}{\sigma\sqrt{t}}$$

1.4.4 Step 4: Comparison with Market Prices

- Compare the theoretical price C to the settle price of the option.
- Calculate error: $\text{Settle Price} - C$
- Visualize both curves on a line graph.

2 Volatility Estimation:

In this section, we estimate the rolling annualized volatility of the stock using historical price data from 2nd May to 1st July. The objective is to compute the parameter σ (volatility) required for the Black-Scholes option pricing model. The process involves the following steps:

- Load necessary R packages and import the dataset containing the closing prices of the stock.
- Compute the natural logarithm of the closing prices, which represents the underlying value of the stock.
- Calculate the daily log returns using the formula: $X_k = \log(P_k) - \log(P_{k-1})$, where P_k is the closing price on day k .
- Remove the first row (as it lacks a prior day for return computation) and append the log return values as a new column in the dataset.
- Compute the rolling average of the log returns over a window size of 59. This window corresponds to the number of business days between 31st January and 2nd May (59 working days).
- Calculate the rolling sample variance of the log returns using the same 59-day window.
- Multiply the rolling standard deviation by $\sqrt{252}$ to scale the volatility to an annual basis, assuming 252 trading days in a year.
- Save the resulting dataset with the newly computed log returns and volatility estimates for subsequent analysis in the option pricing model.

2.1 R Code:

```
# Load necessary packages
library(readr)
library(dplyr)
library(lubridate)

# Read the data
data <- read_csv("C:/Users/UTTATI/OneDrive/Desktop/Projects/DM_project/INFY.NS_data.csv"
)

# View column names and structure
str(data)
head(data)

# Step 1: Take log of each value
log_prices <- log(data$Close)

# Step 2: Compute log returns:  $X_k = \log(P_k) - \log(P_{k-1})$ 
log_returns <- diff(log_prices)

# Step 3: (Optional) add it to our dataset
data_log_returns <- data[-1, ] # Drop first row since diff reduces length by 1
data_log_returns$log_return <- log_returns

# Step 4: View or export
head(data_log_returns) # View first few rows

library(zoo) # For rollmean
# Step 3: Calculate rolling mean of log returns with window size = 59
data_log_returns$rolling_mean_59 <- zoo::rollmean(data_log_returns$log_return, k = 59,
  align = "right", fill = NA)
head(data_log_returns)

# Calculate rolling variance over a window of 59
data_log_returns$rolling_var_59 <- rollapply(
  data_log_returns$log_return,
  width = 59,
  FUN = var,
  align = "right",
  fill = NA
)

# View result
head(data_log_returns, 10)
# Estimate annualized rolling volatility
data_log_returns$rolling_sigma_59 <- sqrt(252 * data_log_returns$rolling_var_59)
head(data_log_returns)
# Save the data_log_returns dataframe to CSV
write_csv(data_log_returns, "data_log_returns.csv", row.names = FALSE)
```

The dataset now includes rolling estimates of daily annualized volatility computed for each trading day between 2nd May and 1st July.

3 Theoretical Option Price Estimation Using the Black–Scholes Model:

Before implementing the Black–Scholes pricing formula, the dataset was enriched by manually adding the column of rolling annualized volatility computed for each trading day from 2nd May to 1st July.

To prepare the data for option pricing, the following preprocessing steps were undertaken:

- Load necessary R packages and import the new dataset containing the date, expiry date, settle price, strike price of the option and volatility (from 2nd May to 1st July).
- The Date and Expiry Date columns were converted to the appropriate Date format for accurate date-based computations.
- The dataset was filtered to include only those records falling within the relevant evaluation window (2nd May to 1st July).

- For each observation, the number of business days (i.e., trading days) between the Date and the corresponding Expiry Date was computed.
- The number of business days to expiry was then converted into time-to-expiry in years by dividing by 252, assuming 252 trading days in a year.
- Define Black- scholes Call and Put option output as function.
For a European call option:

$$C = \frac{S_0 \Phi(\omega) - ke^{-rt} \Phi(\omega - \sigma \sqrt{t})}{\sigma \sqrt{t}}$$

Where

$$\omega = \frac{rt + \sigma^2 t / 2 - \log(k / S_0)}{\sigma \sqrt{t}}$$

For a European put option:

$$P = \frac{Ke^{-rt} \Phi(-\omega + \sigma \sqrt{t}) - S_0 \Phi(-\omega)}{\sigma \sqrt{t}}$$

Where

$$\omega = \frac{rt + \sigma^2 t / 2 - \log(k / S_0)}{\sigma \sqrt{t}}$$

- Theoretical option prices (C for calls and P for puts) were computed for each row, depending on the option type.
- The pricing error was computed as the difference between the market settle price and the theoretical option price—specifically, (*Settle Price* − C) for call options and (*Settle Price* − P) for put options.
- The updated dataset, now containing both theoretical and market option prices, was saved for further analysis and visualization.

3.1 R Code:

```
library(dplyr)
library(lubridate)
library(timedate)
library(readr)
data <- read_csv("C:/Users/UTTATI/OneDrive/Desktop/Projects/DM_project/OPTSTK_INFY_CE_
(1520).csv")

# Convert to Date format
data$Date <- as.Date(data$Date)
data$Expiry <- as.Date(data$Expiry)

# Filter data for target date range
filtered_data <- data %>%
  filter(Date >= as.Date("2025-05-02") & Date <= as.Date("2025-07-01"))

# Calculate business days between DATE and EXPIRY_DATE
filtered_data$working_days <- as.numeric(mapply(function(start, end) {
  sum(isBizday(timeSequence(from = start, to = end, by = "day")))
}, filtered_data$Date, filtered_data$Expiry))

# Convert working days to time to expiry in years
filtered_data$time_to_expiry_years <- filtered_data$working_days / 252

# View result
head(filtered_data)
#Black-Scholes Call Option Price
black_scholes_call <- function(S0, K, r, sigma_hat, t) {
  omega <- (r * t + 0.5 * sigma_hat^2 * t - log(K / S0)) / (sigma_hat * sqrt(t))
  C <- S0 * pnorm(omega) - K * exp(-r * t) * pnorm(omega - sigma_hat * sqrt(t))
  return(C)
}
filtered_data$C <- mapply(black_scholes_call,
  S0 = filtered_data$'Underlying Value',
```

```

        K = filtered_data$'Strike Price',
        r = 0.06,
        sigma_hat = filtered_data$rolling_sigma_59,
        t = filtered_data$time_to_expiry_years)
#To calculate the pricing error
filtered_data$diff <- filtered_data$'Settle Price' - filtered_data$C
# Black-Scholes Put Option Price
black_scholes_put <- function(S0, K, r, sigma_hat, t) {
  omega <- (r * t + 0.5 * sigma_hat^2 * t - log(K / S0)) / (sigma_hat * sqrt(t))
  P <- K * exp(-r * t) * pnorm(-omega + sigma_hat * sqrt(t)) - S0 * pnorm(-omega)
  return(P)
}
filtered_data$P <- mapply(black_scholes_put,
  S0 = filtered_data$'Underlying Value',
  K = filtered_data$'Strike Price',
  r = 0.06,
  sigma_hat = filtered_data$rolling_sigma_59,
  t = filtered_data$time_to_expiry_years)

#To calculate the pricing error
filtered_data$diff <- filtered_data$'Settle Price' - filtered_data$P
#To download the data
write.csv(filtered_data, "filtered_data_with_call_prices.csv", row.names = FALSE)

```

The final dataset has been prepared, containing both the theoretical option prices and the corresponding pricing errors for each observation. Access the final dataset containing all computed values at the following link: [link](#)

4 Plots:

4.1 Theoretical vs Market Option price over time:

For Call option:

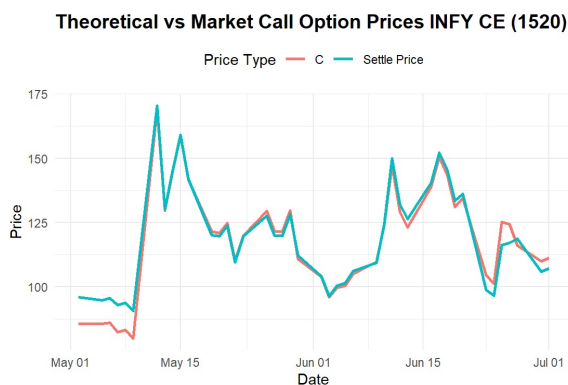


Figure 1: INFY CE (1520)

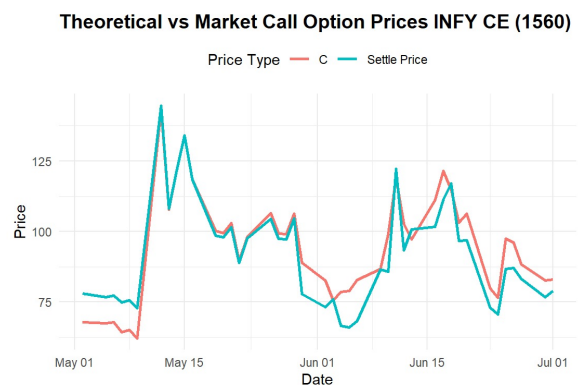


Figure 2: INFY CE (1560)

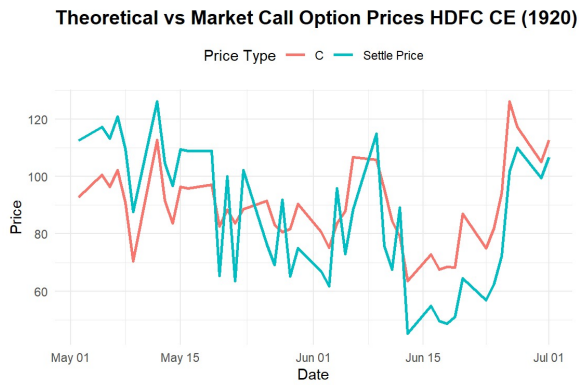


Figure 3: HDFC CE (1920)

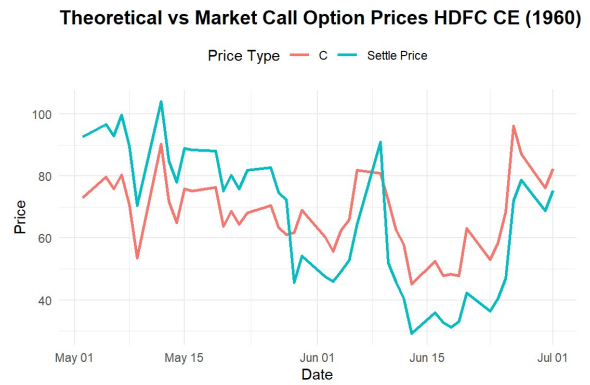


Figure 4: HDFC CE (1960)

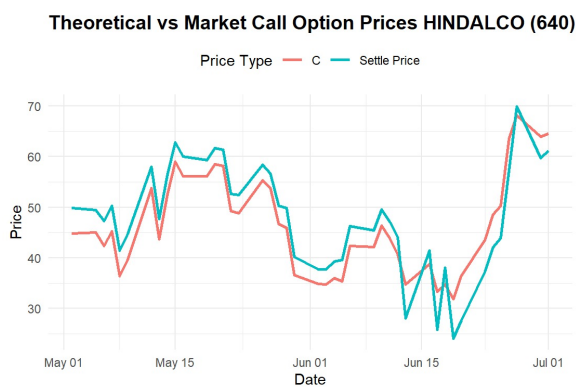


Figure 5: HINDALCO CE (640)

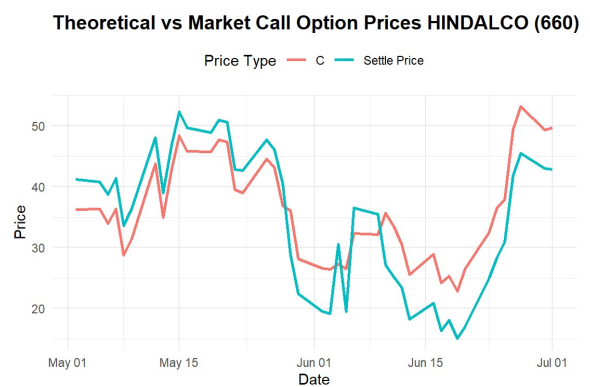


Figure 6: HINDALCO CE (660)

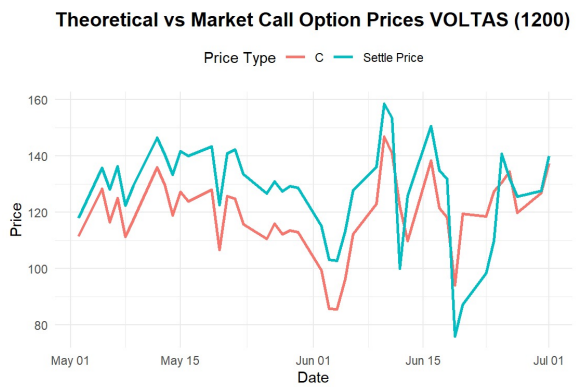


Figure 7: VOLTAS CE (1200)

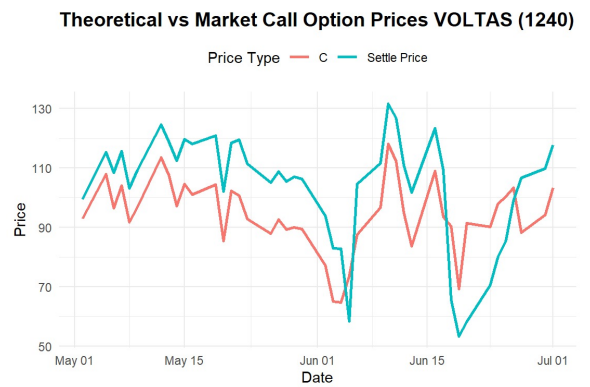


Figure 8: VOLTAS CE (1240)

For Put Option:

Theoretical vs Market Put Option Prices INFY (1520)

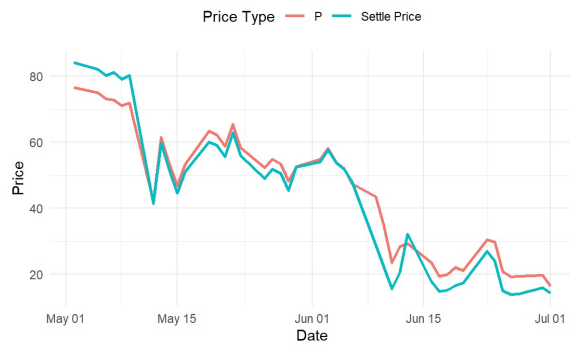


Figure 9: INFY PE (1520)

Theoretical vs Market Put Option Prices INFY (1560)

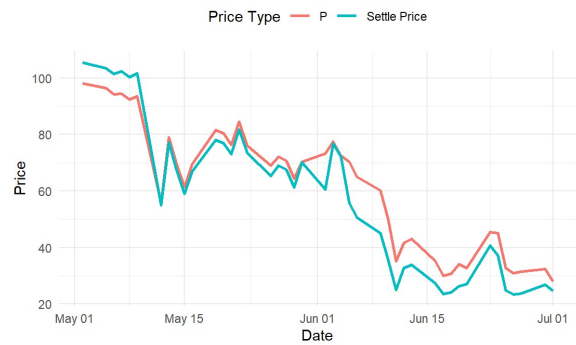


Figure 10: INFY PE (1560)

Theoretical vs Market Put Option Prices HDFC (1920)

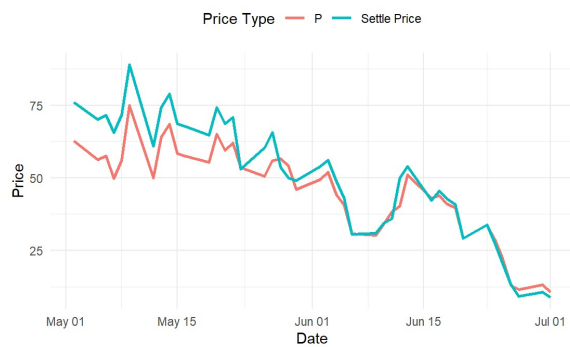


Figure 11: HDFC PE (1920)

Theoretical vs Market Put Option Prices HDFC (1960)

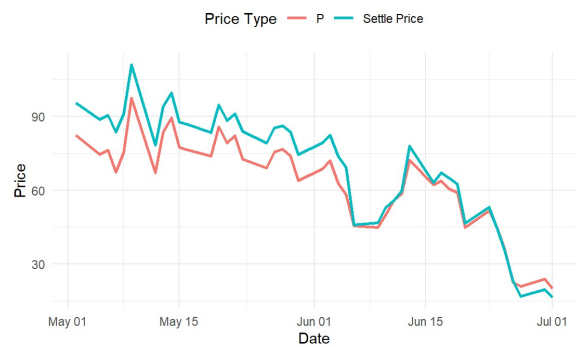


Figure 12: HDFC PE (1960)

Theoretical vs Market Put Option Prices HINDALCO (640)

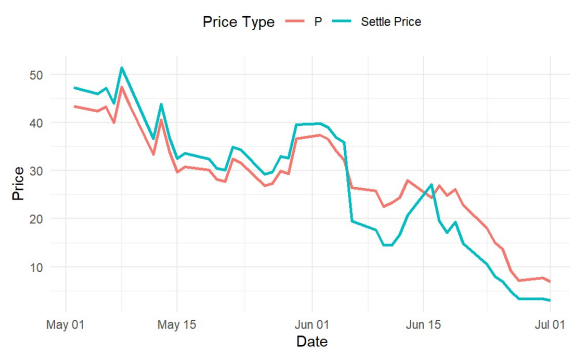


Figure 13: HINDALCO PE (640)

Theoretical vs Market Put Option Prices HINDALCO (660)

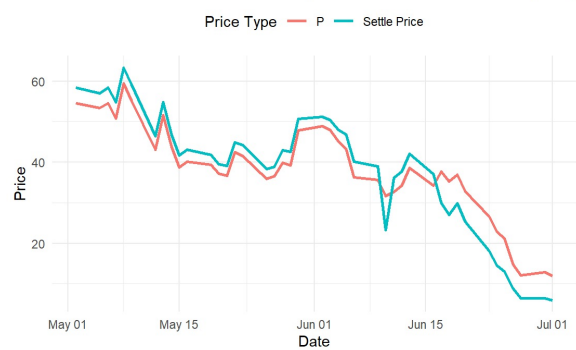


Figure 14: HINDALCO PE (660)

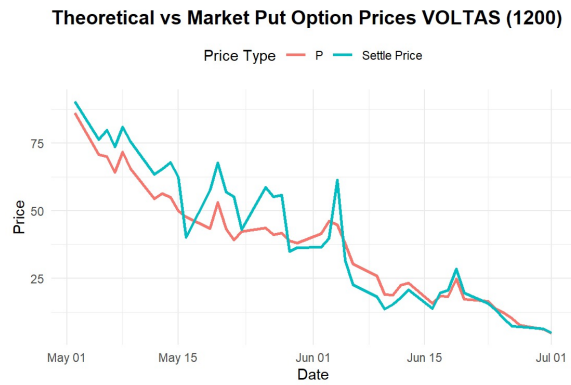


Figure 15: VOLTAS PE (1200)

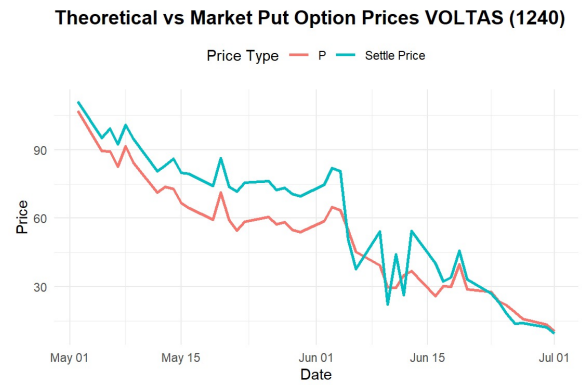


Figure 16: VOLTAS PE (1240)

4.2 Dynamics of Option Pricing Error:

For Call option:



Figure 17: INFY CE (1520)

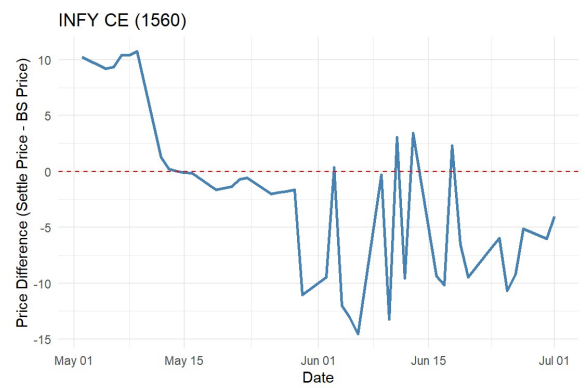


Figure 18: INFY CE (1560)

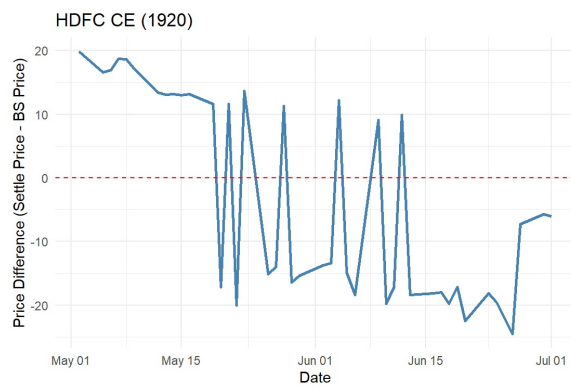


Figure 19: HDFC CE (1920)

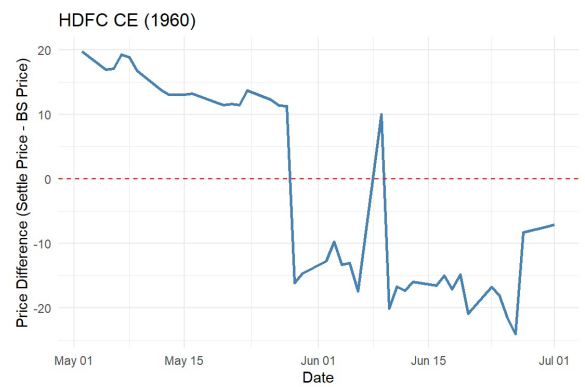


Figure 20: HDFC CE (1960)

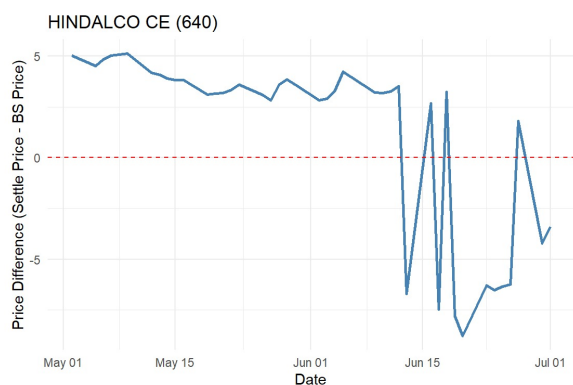


Figure 21: HINDALCO CE (640)

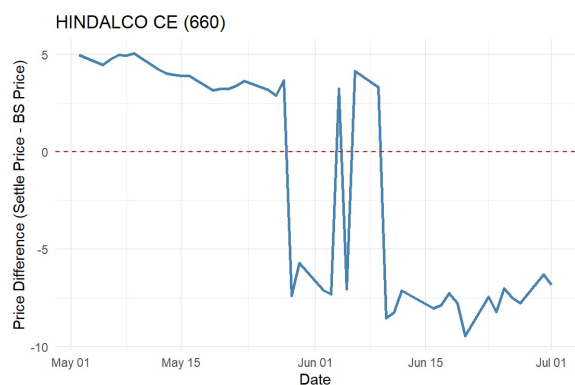


Figure 22: HINDALCO CE (660)



Figure 23: VOLTAS CE (1200)



Figure 24: VOLTAS CE (1240)

For put option:



Figure 25: INFY PE (1520)



Figure 26: INFY PE (1560)

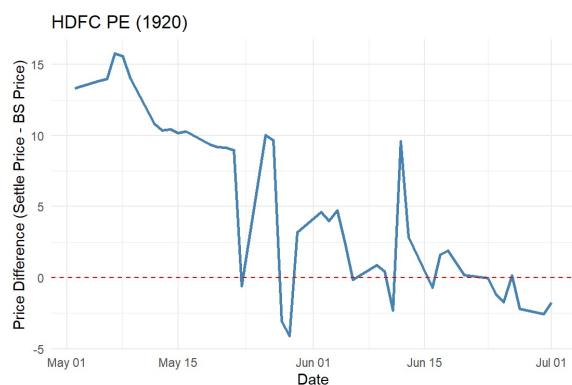


Figure 27: HDFC PE (1920)



Figure 28: HDFC PE (1960)

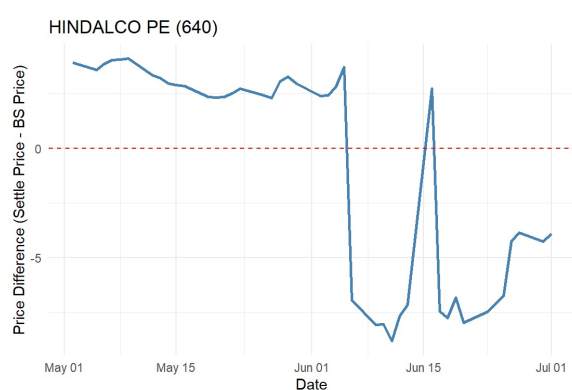


Figure 29: HINDALCO PE (640)

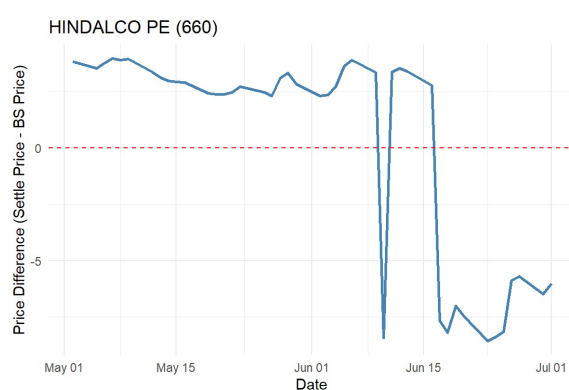


Figure 30: HINDALCO PE (660)

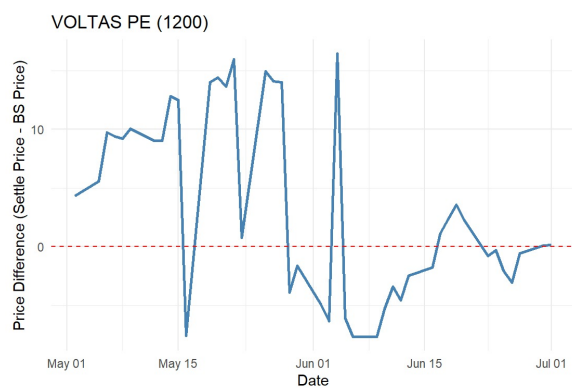


Figure 31: VOLTAS PE (1200)

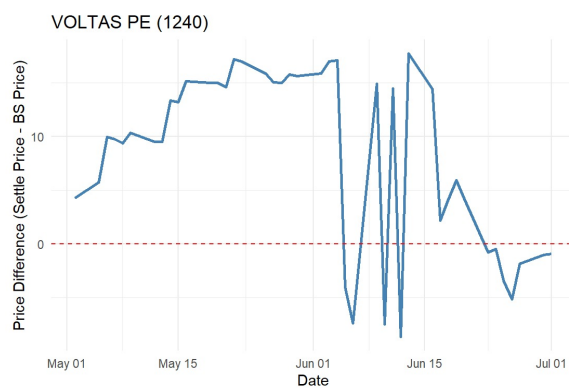


Figure 32: VOLTAS PE (1240)